

Tolvas

Totally Integrated Automation Portal

Block_1 [FB1]

Block_1 Properties

General

Name	Block_1	Number	1	Type	FB	Language	GRAPH
Numbering	Automatic	Network language	LAD	Block version	V2.0		

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
▼ Input									
OFF_SQ	Bool	false	Non-retain	False	False	False	False		Turn sequence off
INIT_SQ	Bool	false	Non-retain	False	False	False	False		Set sequence to initial state
ACK_EF	Bool	false	Non-retain	False	False	False	False		Acknowledge all errors and faults
S_PREV	Bool	false	Non-retain	False	False	False	False		Output previous step in parameter S_NO
S_NEXT	Bool	false	Non-retain	False	False	False	False		Indicate next step in parameter S_NO
SW_AUTO	Bool	false	Non-retain	False	False	False	False		Automatic mode
SW_TAP	Bool	false	Non-retain	False	False	False	False		Semiautomatic/switch with transition
SW_TOP	Bool	false	Non-retain	False	False	False	False		Semiautomatic/ignore transition
SW_MAN	Bool	false	Non-retain	False	False	False	False		Manual mode
S_SEL	Int	0	Non-retain	False	False	False	False		Select step to be output to S_NO
S_ON	Bool	false	Non-retain	False	False	False	False		Activate step indicated in S_NO
S_OFF	Bool	false	Non-retain	False	False	False	False		Disable step indicated in S_NO
T_PUSH	Bool	false	Non-retain	False	False	False	False		Enable transition to switch in semi automatic mode
▼ Output									
S_NO	Int	0	Non-retain	False	False	False	False		Step number
S_MORE	Bool	false	Non-retain	False	False	False	False		More steps are available and can be shown in S_NO
S_ACTIVE	Bool	false	Non-retain	False	False	False	False		Step indicated in S_NO is active
ERR_FLT	Bool	false	Non-retain	False	False	False	False		Interlock or supervision group error
AUTO_ON	Bool	false	Non-retain	False	False	False	False		Automatic mode is active
TAP_ON	Bool	false	Non-retain	False	False	False	False		Semiautomatic mode/step with transition enabled
TOP_ON	Bool	false	Non-retain	False	False	False	False		Semiautomatic mode/ignore transition enabled
MAN_ON	Bool	false	Non-retain	False	False	False	False		Manual mode is active
InOut									
▼ Static									
▼ RT_DATA	G7_RTData-Plus_V2		Non-retain	False	False	False	True		Internal data area
S_DISPLAY	Int	0	Non-retain	False	False	False	False		Internal display of output parameter S_NO
S_SEL_OLD	Int	0	Non-retain	False	False	False	False		Previous value in S_SEL
S_DISPIDX	USInt	255	Non-retain	False	False	False	False		Index of the step in S_NO
T_DISPIDX	USInt	255	Non-retain	False	False	False	False		Index of the transition displayed in T_NO
▼ MOP_EDGE	G7_MOP-Plus_V2		Non-retain	False	False	False	True		Mode in last cycle
AUTO	Bool	false	Non-retain	False	False	False	False		Status: automatic mode
MAN	Bool	false	Non-retain	False	False	False	False		Status: manual mode
TAP	Bool	false	Non-retain	False	False	False	False		Status: semi automatic/switch with transition
TOP	Bool	false	Non-retain	False	False	False	False		Status: semi automatic/ignore transition
ACK_S	Bool	false	Non-retain	False	False	False	False		Request: acknowledge step at parameter S_NO
REG_S	Bool	false	Non-retain	False	False	False	False		Request: register step indicated in S_NO
T_PREV	Bool	false	Non-retain	False	False	False	False		Request: output previous valid transition in T_NO
T_NEXT	Bool	false	Non-retain	False	False	False	False		Request: output next valid transition in T_NO
LOCK	Bool	false	Non-retain	False	False	False	False		Status: interlocks activated
SUP	Bool	false	Non-retain	False	False	False	False		Status: supervisions activated

Totally Integrated Automation Portal													
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision			Comment	
EXEC_BITS[229]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[230]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[231]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[232]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[233]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[234]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[235]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[236]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[237]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[238]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[239]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[240]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[241]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[242]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[243]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[244]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[245]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[246]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[247]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[248]		Bool	false	Non-retain	False	False	False	False				System-internal	
EXEC_BITS[249]		Bool	false	Non-retain	False	False	False	False				System-internal	
▼ OFFSETS		G7_Offsets-Plus_V2		Non-retain	False	False	False	True				Internal offsets	
SINI_OFFSET		UInt	0	Non-retain	False	False	False	False				Offset of internal array SINI[]	
LSTT_OFFSET		UInt	2	Non-retain	False	False	False	False				Offset of internal array LSTT[]	
ATAJ_OFFSET		UInt	25	Non-retain	False	False	False	False				Offset of internal array ATAJ[]	
ATAB_OFFSET		UInt	48	Non-retain	False	False	False	False				Offset of internal array ATAB[]	
PSTT_OFFSET		UInt	71	Non-retain	False	False	False	False				Offset of internal array PSTT[]	
NSTT_OFFSET		UInt	94	Non-retain	False	False	False	False				Offset of internal array NSTT[]	
ASSJ_OFFSET		UInt	117	Non-retain	False	False	False	False				Offset of internal array ASSJ[]	
ASSB_OFFSET		UInt	133	Non-retain	False	False	False	False				Offset of internal array ASSB[]	
PTTS_OFFSET		UInt	149	Non-retain	False	False	False	False				Offset of internal array PTTS[]	
NTTS_OFFSET		UInt	165	Non-retain	False	False	False	False				Offset of internal array NTTS[]	
SW_SQTS_OFFSET		UInt	181	Non-retain	False	False	False	False				Offset of internal array SW_SQTS[]	
SWITCH_OFFSET		UInt	197	Non-retain	False	False	False	False				Offset of internal array SWITCH[]	
TVX_OFFSET		UInt	199	Non-retain	False	False	False	False				Offset of internal array TVX[]	
TTX_OFFSET		UInt	211	Non-retain	False	False	False	False				Offset of internal array TTX[]	
TSX_OFFSET		UInt	223	Non-retain	False	False	False	False				Offset of internal array TSX[]	
S00X_OFFSET		UInt	225	Non-retain	False	False	False	False				Offset of internal array S00X[]	
SOFFX_OFFSET		UInt	227	Non-retain	False	False	False	False				Offset of internal array SOFFX[]	
SONX_OFFSET		UInt	229	Non-retain	False	False	False	False				Offset of internal array SONX[]	
SAX_OFFSET		UInt	231	Non-retain	False	False	False	False				Offset of internal array SAX[]	
SERRX_OFFSET		UInt	233	Non-retain	False	False	False	False				Offset of internal array SERRX[]	
SMX_OFFSET		UInt	250	Non-retain	False	False	False	False				Offset of internal array SMX[]	
SOX_OFFSET		UInt	267	Non-retain	False	False	False	False				Offset of internal array SOX[]	
S1X_OFFSET		UInt	284	Non-retain	False	False	False	False				Offset of internal array S1X[]	
THRESHOLD_SUP		USInt	0	Non-retain	False	False	False	False				Threshold for step activation time	
THRESHOLD_WARN		USInt	0	Non-retain	False	False	False	False				Threshold for step activation time (warning only)	
▼ Trans1		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	1	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans2		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	

Totally Integrated Automation Portal													
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision			Comment	
TNO		Int	13	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans14		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	14	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans15		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	15	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans16		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	16	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans17		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	17	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans18		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	18	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans19		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	19	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans20		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	20	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans21		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	21	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans22		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	22	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Trans23		G7_Transition-Plus_V2		Non-retain	False	False	False	True				Transition structure	
TV		Bool	false	Non-retain	False	False	False	False				Transition is valid	
TT		Bool	false	Non-retain	False	False	False	False				Transition is satisfied	
TS		Bool	false	Non-retain	False	False	False	False				Transition switches	
TNO		Int	23	Non-retain	False	False	False	False				Indicates the user-defined transition number	
▼ Step1		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure	
S1		Bool	false	Non-retain	False	False	False	False				Step is activated	
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state	
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state	
R1		Bool	false	Non-retain	False	False	False	False				Reserved	
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged	
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated	
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state	

Totally Integrated Automation Portal												
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision			Comment
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state
X		Bool	false	Non-retain	False	False	False	False				Step is active
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied
VA		Bool	false	Non-retain	False	False	False	False				Supervision active
RA		Bool	false	Non-retain	False	False	False	False				Reserved
AA		Bool	false	Non-retain	False	False	False	False				Reserved
SS		Bool	false	Non-retain	False	False	False	False				System-internal
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision
SNO		Int	1	Non-retain	False	False	False	False				User step number
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time
SM		Bool	false	Non-retain	False	False	False	False				System-internal
H_IL_ERR		Byte	16#0	Non-retain	False	False	False	False				System-internal
H_SV_FLT		Byte	16#04	Non-retain	False	False	False	False				System-internal
▼ Step2		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure
S1		Bool	false	Non-retain	False	False	False	False				Step is activated
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state
R1		Bool	false	Non-retain	False	False	False	False				Reserved
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state
X		Bool	false	Non-retain	False	False	False	False				Step is active
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied
VA		Bool	false	Non-retain	False	False	False	False				Supervision active
RA		Bool	false	Non-retain	False	False	False	False				Reserved
AA		Bool	false	Non-retain	False	False	False	False				Reserved
SS		Bool	false	Non-retain	False	False	False	False				System-internal
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision
SNO		Int	2	Non-retain	False	False	False	False				User step number
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time
SM		Bool	false	Non-retain	False	False	False	False				System-internal
H_IL_ERR		Byte	16#0	Non-retain	False	False	False	False				System-internal
H_SV_FLT		Byte	16#04	Non-retain	False	False	False	False				System-internal
▼ Step3		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure
S1		Bool	false	Non-retain	False	False	False	False				Step is activated
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state
R1		Bool	false	Non-retain	False	False	False	False				Reserved
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state
X		Bool	false	Non-retain	False	False	False	False				Step is active
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied
VA		Bool	false	Non-retain	False	False	False	False				Supervision active
RA		Bool	false	Non-retain	False	False	False	False				Reserved
AA		Bool	false	Non-retain	False	False	False	False				Reserved
SS		Bool	false	Non-retain	False	False	False	False				System-internal
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision
SNO		Int	3	Non-retain	False	False	False	False				User step number
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time

Totally Integrated Automation Portal													
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision			Comment	
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state	
R1		Bool	false	Non-retain	False	False	False	False				Reserved	
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged	
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated	
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state	
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state	
X		Bool	false	Non-retain	False	False	False	False				Step is active	
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied	
VA		Bool	false	Non-retain	False	False	False	False				Supervision active	
RA		Bool	false	Non-retain	False	False	False	False				Reserved	
AA		Bool	false	Non-retain	False	False	False	False				Reserved	
SS		Bool	false	Non-retain	False	False	False	False				System-internal	
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock	
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision	
SNO		Int	9	Non-retain	False	False	False	False				User step number	
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time	
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance	
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time	
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time	
SM		Bool	false	Non-retain	False	False	False	False				System-internal	
H_IL_ERR		Byte	16#0	Non-retain	False	False	False	False				System-internal	
H_SV_FLT		Byte	16#04	Non-retain	False	False	False	False				System-internal	
▼ Step10		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure	
S1		Bool	false	Non-retain	False	False	False	False				Step is activated	
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state	
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state	
R1		Bool	false	Non-retain	False	False	False	False				Reserved	
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged	
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated	
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state	
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state	
X		Bool	false	Non-retain	False	False	False	False				Step is active	
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied	
VA		Bool	false	Non-retain	False	False	False	False				Supervision active	
RA		Bool	false	Non-retain	False	False	False	False				Reserved	
AA		Bool	false	Non-retain	False	False	False	False				Reserved	
SS		Bool	false	Non-retain	False	False	False	False				System-internal	
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock	
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision	
SNO		Int	10	Non-retain	False	False	False	False				User step number	
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time	
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance	
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time	
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time	
SM		Bool	false	Non-retain	False	False	False	False				System-internal	
H_IL_ERR		Byte	16#0	Non-retain	False	False	False	False				System-internal	
H_SV_FLT		Byte	16#04	Non-retain	False	False	False	False				System-internal	
▼ Step11		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure	
S1		Bool	false	Non-retain	False	False	False	False				Step is activated	
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state	
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state	
R1		Bool	false	Non-retain	False	False	False	False				Reserved	
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged	
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated	
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state	
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state	
X		Bool	false	Non-retain	False	False	False	False				Step is active	
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied	
VA		Bool	false	Non-retain	False	False	False	False				Supervision active	
RA		Bool	false	Non-retain	False	False	False	False				Reserved	
AA		Bool	false	Non-retain	False	False	False	False				Reserved	
SS		Bool	false	Non-retain	False	False	False	False				System-internal	
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock	
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision	
SNO		Int	11	Non-retain	False	False	False	False				User step number	
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time	

Totally Integrated Automation Portal													
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision			Comment	
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance	
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time	
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time	
SM		Bool	false	Non-retain	False	False	False	False				System-internal	
H_IL_ERR		Byte	16#0	Non-retain	False	False	False	False				System-internal	
H_SV_FLT		Byte	16#04	Non-retain	False	False	False	False				System-internal	
▼ Step12		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure	
S1		Bool	false	Non-retain	False	False	False	False				Step is activated	
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state	
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state	
R1		Bool	false	Non-retain	False	False	False	False				Reserved	
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged	
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated	
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state	
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state	
X		Bool	false	Non-retain	False	False	False	False				Step is active	
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied	
VA		Bool	false	Non-retain	False	False	False	False				Supervision active	
RA		Bool	false	Non-retain	False	False	False	False				Reserved	
AA		Bool	false	Non-retain	False	False	False	False				Reserved	
SS		Bool	false	Non-retain	False	False	False	False				System-internal	
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock	
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision	
SNO		Int	12	Non-retain	False	False	False	False				User step number	
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time	
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance	
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time	
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time	
SM		Bool	false	Non-retain	False	False	False	False				System-internal	
H_IL_ERR		Byte	16#0	Non-retain	False	False	False	False				System-internal	
H_SV_FLT		Byte	16#04	Non-retain	False	False	False	False				System-internal	
▼ Step13		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure	
S1		Bool	false	Non-retain	False	False	False	False				Step is activated	
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state	
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state	
R1		Bool	false	Non-retain	False	False	False	False				Reserved	
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged	
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated	
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state	
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state	
X		Bool	false	Non-retain	False	False	False	False				Step is active	
LA		Bool	false	Non-retain	False	False	False	False				Interlock is not satisfied	
VA		Bool	false	Non-retain	False	False	False	False				Supervision active	
RA		Bool	false	Non-retain	False	False	False	False				Reserved	
AA		Bool	false	Non-retain	False	False	False	False				Reserved	
SS		Bool	false	Non-retain	False	False	False	False				System-internal	
LS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed interlock	
VS		Bool	false	Non-retain	False	False	False	False				Direct result of the programmed supervision	
SNO		Int	13	Non-retain	False	False	False	False				User step number	
T		Time	T#0ms	Non-retain	False	False	False	False				Total step activation time	
U		Time	T#0ms	Non-retain	False	False	False	False				Step activation time without disturbance	
T_MAX		Time	T#10S	Non-retain	False	False	False	False				Maximal step activation time	
T_WARN		Time	T#7S	Non-retain	False	False	False	False				Warning time	
SM		Bool	false	Non-retain	False	False	False	False				System-internal	
H_IL_ERR		Byte	16#0	Non-retain	False	False	False	False				System-internal	
H_SV_FLT		Byte	16#04	Non-retain	False	False	False	False				System-internal	
▼ Step14		G7_Step-Plus_V2		Non-retain	False	False	False	True				Step structure	
S1		Bool	false	Non-retain	False	False	False	False				Step is activated	
L1		Bool	false	Non-retain	False	False	False	False				interlock leaving state	
V1		Bool	false	Non-retain	False	False	False	False				Supervision entering state	
R1		Bool	false	Non-retain	False	False	False	False				Reserved	
A1		Bool	false	Non-retain	False	False	False	False				Error is acknowledged	
S0		Bool	false	Non-retain	False	False	False	False				Step is deactivated	
L0		Bool	false	Non-retain	False	False	False	False				Interlock entering state	
V0		Bool	false	Non-retain	False	False	False	False				Supervision leaving state	

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False		System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False		System-internal
Temp									
Constant									

Alarms

Enable alarms			True		
---------------	--	--	------	--	--

Category	Category enabler	Display class
Error		0
Warning		0
Info		0
Category 4		0
Category 5		0
Category 6		0
Category 7		0
Category 8		0

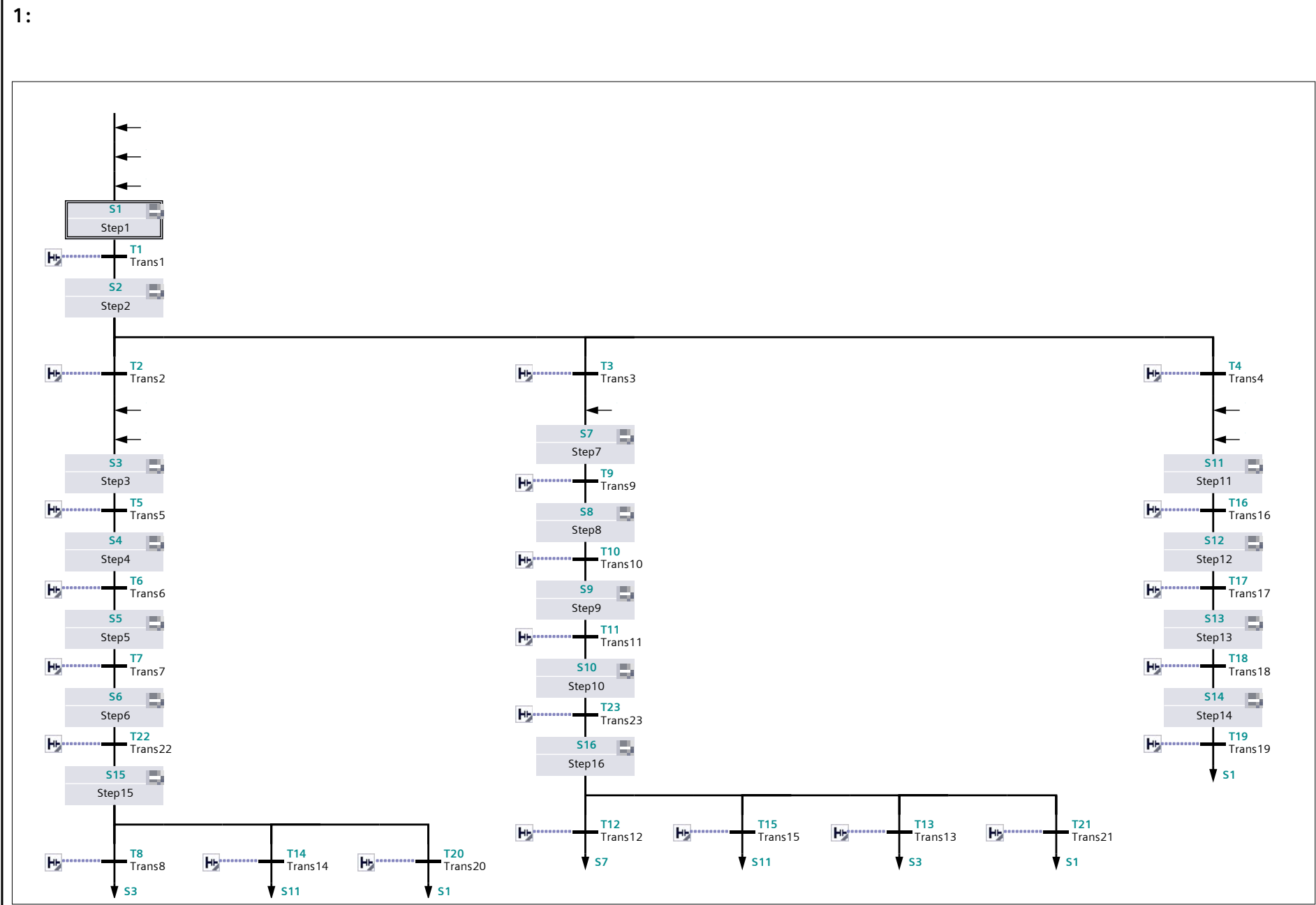
Category for interlocks	Error	Subcategory 1 for interlocks		Subcategory 2 for interlocks	
Category for supervisions	Error	Subcategory 1 for supervisions		Subcategory 2 for supervisions	
Category for GRAPH warnings	Warning	Subcategory 1 for GRAPH warnings		Subcategory 2 for GRAPH warnings	

Symbol	Address	Type	Comment
--------	---------	------	---------

Permanent pre-instructions

1:

Sequences (1)



S1 - [Initial step]:Step1

Totally Integrated Automation Portal

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		N	"i_T500":=0
		N	"i_T600":=0
		R	"b_M_in"
		R	"b_M_out"
		R	"b_Piston_A"
		R	"b_Piston_B"

T1:Trans1

%M0.0

"b_Start"

%MW12

"i_Peso_Des"

>

Int

0

S2:Step2

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

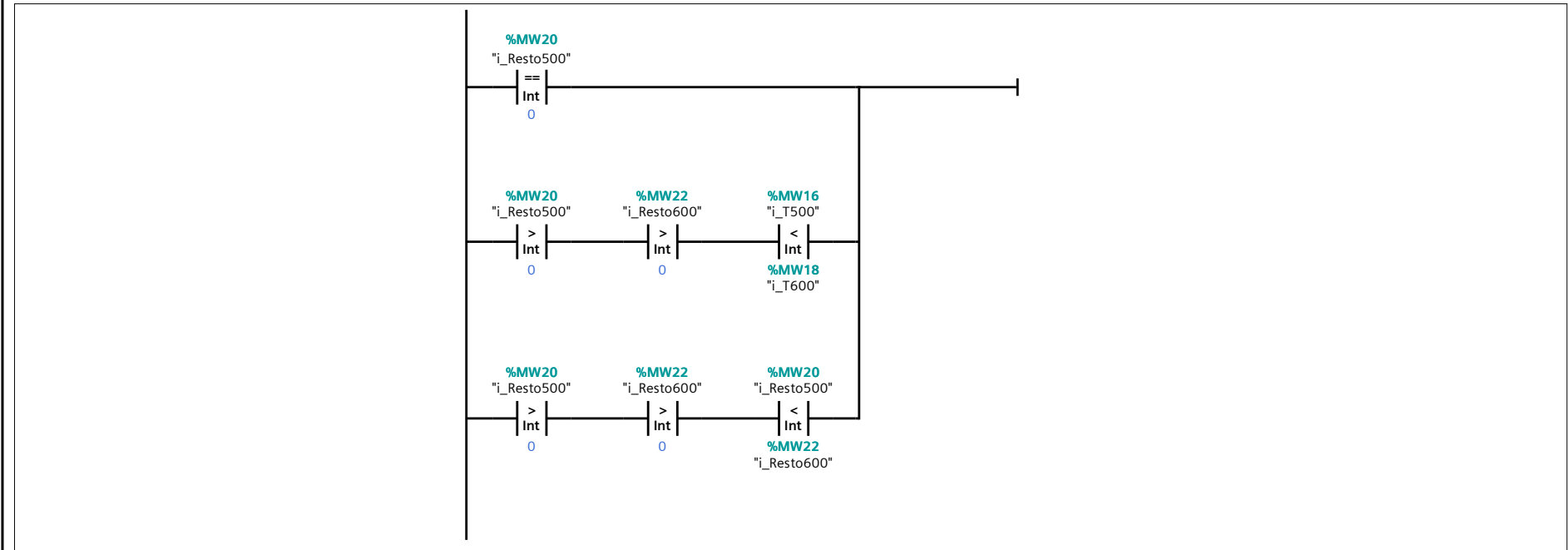
(v)

Actions:

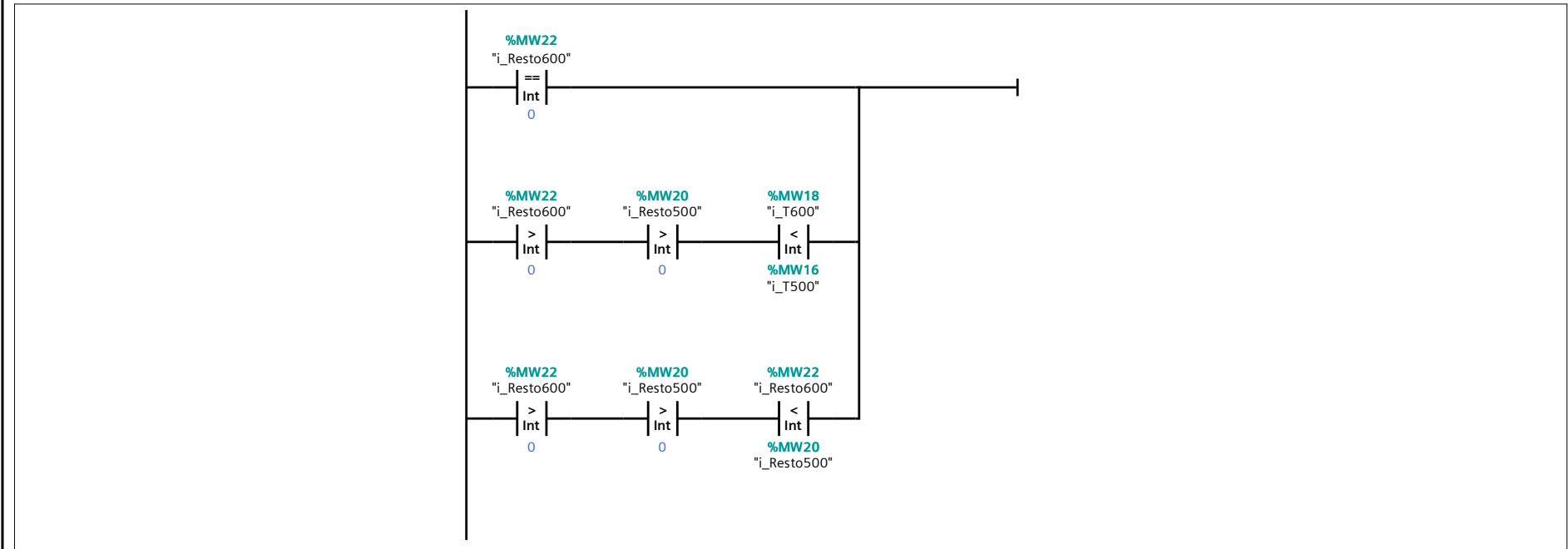
Actions:

Interlock	Event	Qualifier	Action
		N	"i_T500":= "i_Peso_Des" / 500
		N	"i_T600":= "i_Peso_Des" / 600
		N	"i_Resto500":= "i_Peso_Des" % 500
		N	"i_Resto600":= "i_Peso_Des" % 600

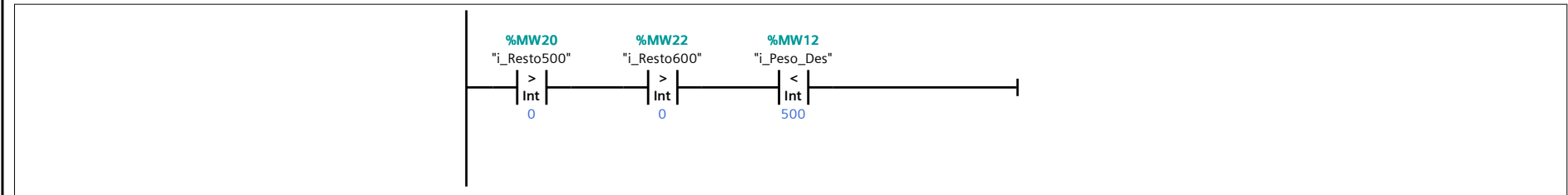
T2:Trans2



T3:Trans3

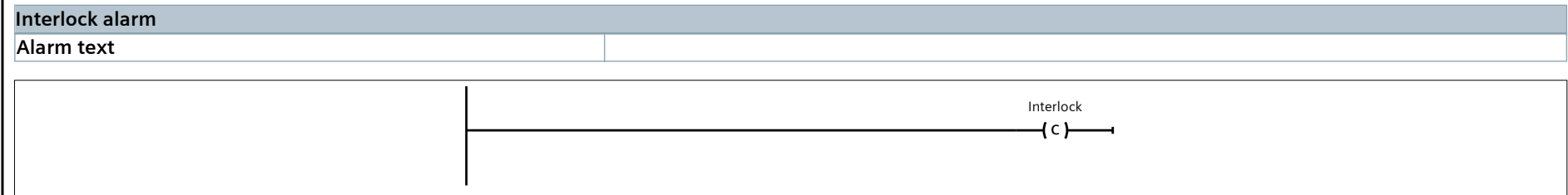


T4:Trans4

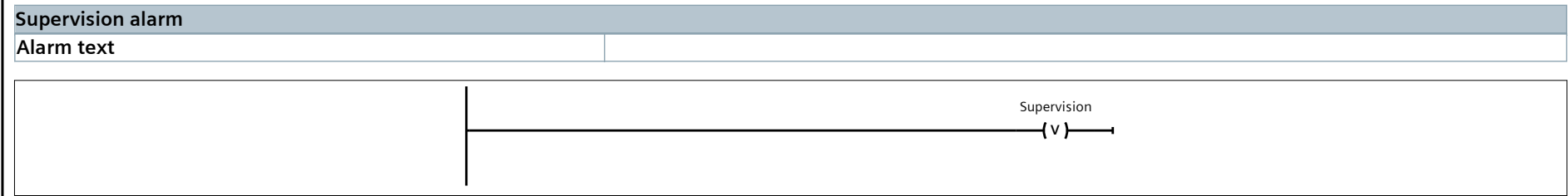


S3:Step3

Interlock -(c)-:



Supervision -(v)-:



Actions:

Actions:			
Interlock	Event	Qualifier	Action
		S	"b_Piston_A"
		R	"b_Piston_B"
		R	"b_M_out"
		R	"b_TempS"

Totally Integrated Automation Portal

T5:Trans5

%M0.1

"b_A_ext"

%M0.4

"b_B_ret"

S4:Step4

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"b_M_in"

T6:Trans6

%M0.5

"b_T500_Full"

S5:Step5

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		R	"b_M_in"
		R	"b_Piston_A"

T7:Trans7

%M0.2

"b_A_ret"

%M0.6

"b_T500_Emp"

S6:Step6

Totally Integrated Automation Portal

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"b_M_out"
		S	"b_TempS"

T22:Trans22

%M1.6
"b_TempF"

S7:Step7

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		R	"b_Piston_A"
		S	"b_Piston_B"
		R	"b_M_out"
		R	"b_TempS"

T9:Trans9

%M0.2
"b_A_ret"

%M0.3
"b_B_ext"

S8:Step8

Interlock -(c)-:

Interlock alarm

Alarm text

Totally Integrated Automation Portal		
--------------------------------------	--	--

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"b_M_in"

T10:Trans10

%M0.7

"b_T600_Full"

S9:Step9

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		R	"b_M_in"
		R	"b_Piston_B"

T11:Trans11

%M0.4

"b_B_ret"

%M1.0

"b_T600_Emp"

S10:Step10

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

--	--	--

Totally Integrated Automation Portal

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"b_M_out"
		S	"b_TempS"

T23:Trans23

%M1.6

"b_TempF"

S11:Step11

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		R	"b_Piston_A"
		S	"b_Piston_B"
		R	"b_M_out"
		R	"b_TempS"

T16:Trans16

%M0.2

"b_A_ret"

%M0.3

"b_B_ext"

S12:Step12

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"b_M_in"
		S	"b_TempS"

Totally Integrated Automation Portal		
--------------------------------------	--	--

Interlock	Event	Qualifier	Action
		N	"i_Peso_Des" := 0

T19:Trans19

%M1.6

"b_TempF"

S15:Step15

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		N	"i_Peso_Des" := "i_Peso_Des" - 500

T8:Trans8

%MW12

"i_Peso_Des"

>=

Int

500

T14:Trans14

%MW12

"i_Peso_Des"

<

Int

500

%MW12

"i_Peso_Des"

>

Int

0

T20:Trans20

%MW12

"i_Peso_Des"

==

Int

0

S16:Step16

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Totally Integrated Automation Portal		
--------------------------------------	--	--

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		N	"i_Peso_Des" := "i_Peso_Des" - 600

T12:Trans12

%MW12

"i_Peso_Des"

>=

Int

600

T15:Trans15

%MW12

"i_Peso_Des"

<

Int

500

%MW12

"i_Peso_Des"

>

Int

0

T13:Trans13

%MW12

"i_Peso_Des"

<

Int

600

%MW12

"i_Peso_Des"

>=

Int

500

T21:Trans21

%MW12

"i_Peso_Des"

=

Int

0

Permanent post-instructions

1: